[T]-Theory: Computing

P7 · Conscious Hardware · For: The Lean 4 Engineer

Affective State Propagator — the proof tree and its open obligations.

Build state (2026-08-15):

- `lake build`: **3912/3912**, exit 0
- Lean 4.31.0 + Mathlib v4.31.0 + physlib v4.31.0
- Float: **0** in proof files (ISS-009 closed)
- Sorry: **2** bare (Hopfield, ISS-011)

22 defaultTargets:

- SomaField, Hopfield, BRECVEMAField/Variational
- CosmologicalConstant, G2Compactification
- LocalGR, LocalGeometry (new, 2026-08-15)
- UniversalSomaticField, ScaleUniverse

I · Gate Theorem Architecture

The proof tree has two local gate files that discharge vacuous axioms:

LocalGR.lean — Linearised GR gate:

- `RigidAttractor`: strict CY minimum (StrictConvexOn)
- `g2_implies_omega_lambda_static`: PROVED
- Blocks: `g2_holonomy_implies_rigid_attractor` (Berger, pending Mathlib)

LocalGeometry.lean — Compact geometry gate:

- `dm_gauge_neutral`: `gaugeCoupling A = 0 := rfl`
- `brane_localisation_from_g2`: PROVED
- Blocks: `g2_implies_hw_compactification` (KK spectral theory)

II · Key Proved Theorems

`greens_fn_is_SHO`	trivial	Problem 1 closed
`X7_is_7D_product`	trivial	Problem 2 closed
`calabi_yau_moduli_static`	LocalGR	$d\Omega_\Lambda/dz = 0$
`dm_gauge_coupling_zero`	rfl	Product manifold
`usf_all_predictions`	norm_num	Planck 2018
`brecvema_G2_decomp`	simp	$\mathrm{tr}(W_8 - \frac{6}{5}I)$=0

III · Affective Computing: The G-ID

The USF as a computational substrate:

$$\text{Agent field} \leftarrow G_{\text{affective}} \cdot J_{\text{stimulus}}$$

The **Affective State Propagator** maps external stimuli to internal BRECVEMA field states. In multi-agent systems (P19), the Green's function propagates coordination signals between agents in a single step.

IV · ISS Backlog

- **ISS-005**: `brainStemActivatesContagion` — needs computable $W_8^{\mathbb{Q}}$
- **ISS-011**: Hopfield SpinState upgrade for convergence proofs
- **ISS-017**: Auto-regen lean-appendix in release-check
- **Mathlib gap**: Riemannian holonomy · GR perturbation theory · KK spectral theory

G-ID: *Affective State Propagator* — computational kernel for agent field dynamics. ORCID: 0009-0007-2194-0850 · · Zurich 2026